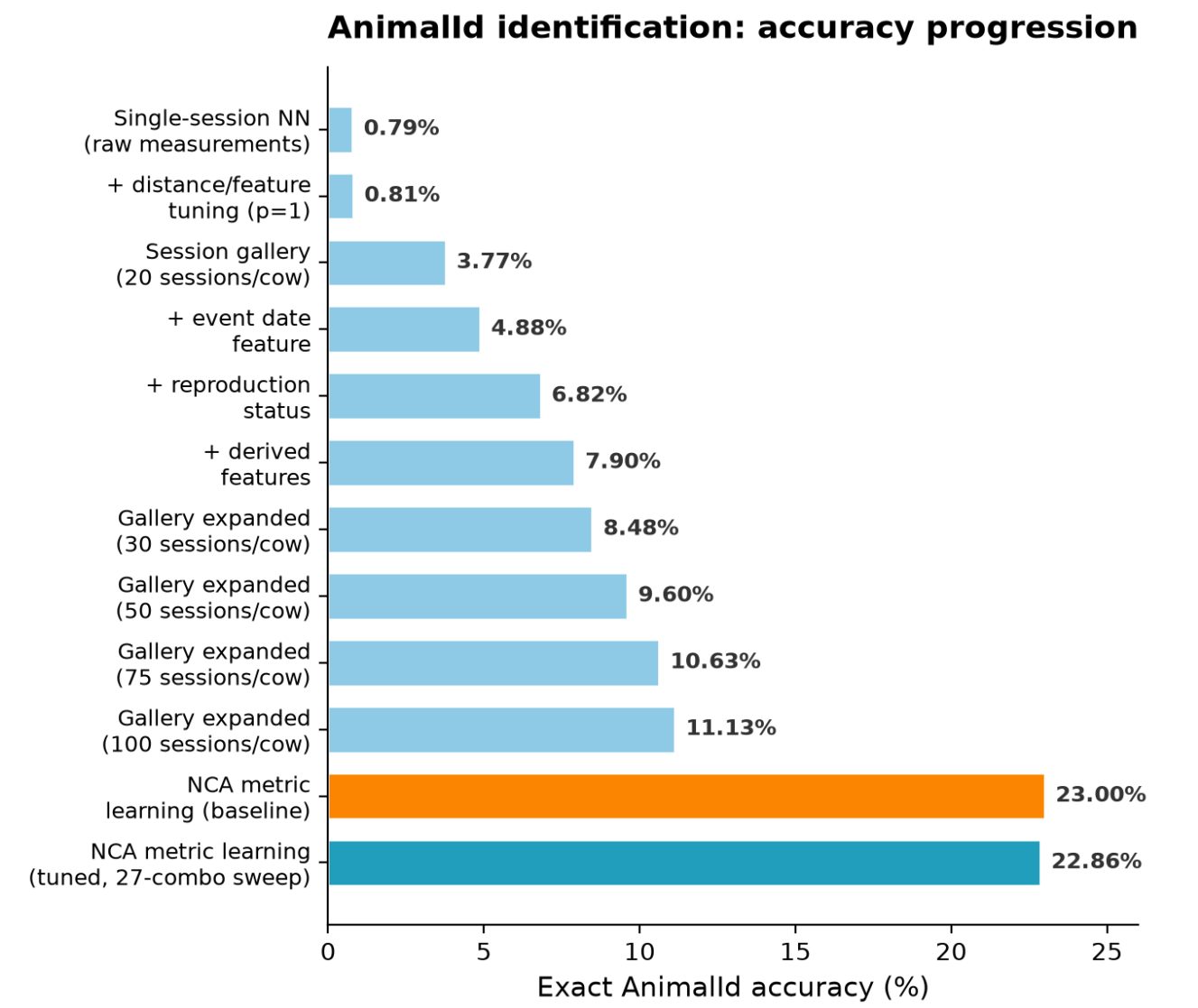


Tuning the NCA Metric-Learning Model

AnimalId Identification Pipeline | ANSCI 6040 Dairy Session Data | 8.5M rows

Standardized Nearest Neighbors (baseline)
18% exact | 39% top-5

NCA Metric Learning (tuned, best of 27)
23% exact | 52% top-5



Expanded NCA Hyperparameter Sweep (27 configurations)

Init strategy	Best config	Exact acc.	Top-5 cov.
auto	n=6, iter=100	23%	52%
identity	n=6, iter=100	20%	48%
pca	n=8, iter=100	19%	48%

Key Takeaways

- ▶ NCA metric learning nearly doubles exact-match accuracy over the standardized-distance baseline (18% → 23%) and raises top-5 coverage from 39% to 52%.
- ▶ init="auto" dominates every configuration tested; "pca" and "identity" initializations underperform across the board.
- ▶ max_iter has little effect once init="auto" is used — 50 iterations already matches 300.
- ▶ Reducing n_components below 6 hurts accuracy (4-component configs drop to 15–19%).
- ▶ Re-validated the full 49-cell / 8.5M-row notebook end-to-end after each change — zero regressions.